

Operating Systems for Intelligent Systems (ENGS2306)

Assignment 5

Bash Scripting Basics

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1. Explore Commandes:

1. `#!/bin/bash`

Explanation: Specifies that the script should be run using the Bash shell.

2. `read -p "Please enter the username: " username`

Explanation: Prompts the user to enter a username and stores it in the `username` variable.`

3. `if id -u "$username" >/dev/null 2>&1; then`

Explanation: Checks if the user exists by trying to get their user ID. The output is redirected to `/dev/null` to suppress it.

4. `sudo useradd -m "$username"`

Explanation: Creates a new user with a home directory (`-m` flag).

5. `sudo mkdir -p "/srv/project$username"`

Explanation: Creates a project directory for the user in `/srv/`.

6. `sudo chown "$username":"$username" "/srv/project$username"`

Explanation: Changes ownership of the directory to the new user.

7. `echo "Successfully created..."`

Explanation: Prints a success message.

*The complete code of your `create_project_user.sh` script (copy-paste as text)

```
#!/bin/bash
```

```
# Prompt for username
```

```
read -p "Please Enter the username: " username
```

```
# Check if user exists
```

```
if id -u "$username" > /dev/null 2>&1; then
```

```
    echo "Error: User $username already exists!"
```

```
    exit 1
```

```
else
```

```
    # Create user with home directory
```

```
    sudo useradd -m "$username"
```

```
    # Create project directory
```

```
    project_dir="/srv/project$username"
```

```
    sudo mkdir -p "$project_dir"
```

```
    # Change ownership
```

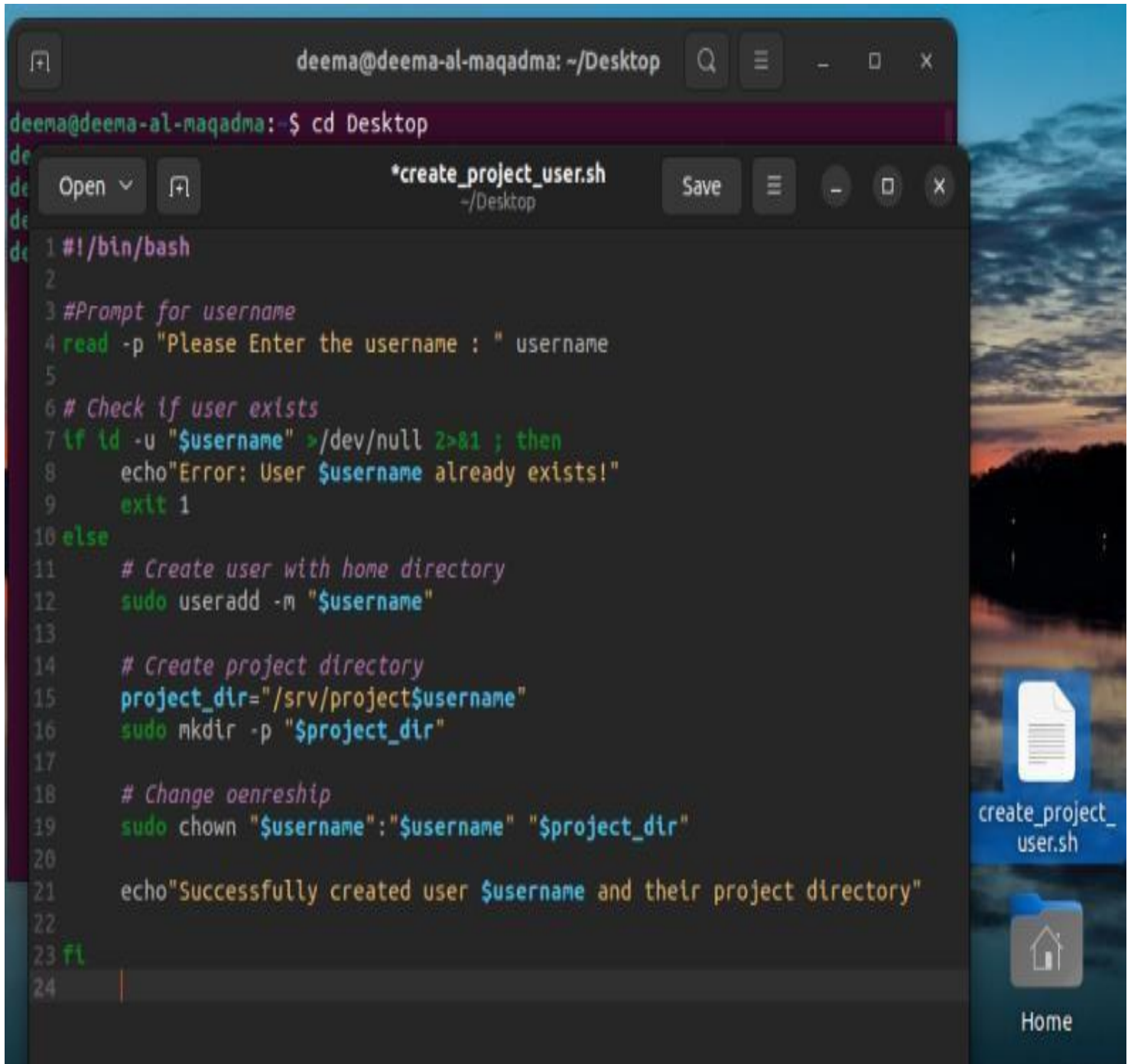
```
    sudo chown "$username":"$username" "$project_dir"
```

```
    echo "Successfully created user $username and their project directory"
```

```
fi
```

* A screenshot showing the script :

(create_project_user.sh script)



The screenshot shows a terminal window with the following content:

```
deema@deema-al-maqadma: ~/Desktop
deema@deema-al-maqadma:~$ cd Desktop
deema@deema-al-maqadma:~/Desktop$ ./create_project_user.sh
```

The script content is as follows:

```
1 #!/bin/bash
2
3 #Prompt for username
4 read -p "Please Enter the username : " username
5
6 # Check if user exists
7 if id -u "$username" >/dev/null 2>&1 ; then
8     echo"Error: User $username already exists!"
9     exit 1
10 else
11     # Create user with home directory
12     sudo useradd -m "$username"
13
14     # Create project directory
15     project_dir="/srv/project$username"
16     sudo mkdir -p "$project_dir"
17
18     # Change ownership
19     sudo chown "$username":"$username" "$project_dir"
20
21     echo"Successfully created user $username and their project directory"
22
23 fi
```

The terminal window also shows a file manager icon for 'create_project_user.sh' and a 'Home' button on the right side.

- * A screenshot showing the script running successfully for a new user.
- A second screenshot showing the script failing gracefully when you try to create the same user again.

```

deema@deema-al-maqadma:~$ cd Desktop
deema@deema-al-maqadma:~/Desktop$ touch create_project_user.sh
deema@deema-al-maqadma:~/Desktop$ chmod +x create_project_user.sh
deema@deema-al-maqadma:~/Desktop$ ./create_project_user.sh

```

```

deema@deema-al-maqadma:~/Desktop$ ./create_project_user.sh
Please Enter the username : Mohammed
Successfully created user Mohammed and their project directory
deema@deema-al-maqadma:~/Desktop$ ./create_project_user.sh
Please Enter the username : Mohammed
Error: User Mohammed already exists!
deema@deema-al-maqadma:~/Desktop$

```

9. Report Explanation :

- > The `id -u "$username"` command checks if the user exists. If it returns exit code 0, the user exists, and the script exits with an error.
- > `$?` stores the exit code of the last command. If it's not 0, the user doesn't exist, and the script proceeds to create them.

" واخر دعواهم ان الحمد لله رب العالمين "